

ITAI7.110 Machine Learning Project Assessment

AIM OF THE PROJECT ASSESSMENT

In this assessment, you need to design and develop solutions for Data Analysis using Machine Learning methods and tools.

The learning outcomes to achieve in this project is:

1. Explain Machine Learning and Artificial Intelligence theories and their impact on society.
2. Design a solution to solve a real-world scenario using Machine Learning and Artificial Intelligence theories.
3. Apply Machine Learning and Artificial Intelligence theories to a real-world scenario.

PROJECT OVERVIEW

This project will consist of a series of tasks to be completed in your own time. It is to be completed using the Python languages and various techniques learnt in the course using the datasets that are mentioned for each task.

DELIVERABLES

For each task, you will create **one ZIP Solution file**. Please put everything in one folder. Once you have solved all four tasks, then please ZIP the solution folder. You need to submit this ZIP folder on Zoodle. Please make sure that your Python code is working straight out of the ZIP file. You may be asked to meet your lecturer to explain your code and answer questions from your lecturer, in case of doubts.

SUBMISSION

You will find a Zoodle submission link to submit your ZIP Solution Folder. You have to submit your work by the due date and time provided.

TASK 1:

LINEAR REGRESSION – CASE STUDY: FINANCIAL DATA ANALYSIS AND VISUALIZATION

QUANDL library provides stock exchange data that can be used to predict the future values of the stocks. It provides us the data according to the stock symbols. More details on QUANDL are on the following URL:

<https://www.quandl.com>

First you need to register yourself on this website and get your personal API-KEY to access data from QUANDL databases. For detailed documentation, please see URL:

<https://docs.quandl.com>

How to configure it with Python and use it in your analysis, please see URL:

<https://docs.quandl.com/docs/python-installation>

You can find a list of Symbols from WIKI Prices (WIKIP) on the following link:

<https://www.quandl.com/databases/WIKIP>

To install QUANDL library, run the following command:

```
pip install quandl
```

Using the get(SYMBOL) method, we can read data from WIKIP in QUANDL, where SYMBOL is a string that represents the stock symbol of a company. Using the following commands, we can read a dataset using QUANDL:

```
import pandas as pd
import quandl as qnd
# Symbol of the stock that we want for analysis
symbol = "WIKI/AAPL"
qnd.ApiConfig.api_key = "#####" # you will get your KEY
df = qnd.get(symbol)
print(df.info())
```

Here in this example, we are reading APPLE's stock data in the "df" variable as a PANDAS data frame. It contains 12 columns as shown below:

```
Data columns (total 12 columns):
Open          9400 non-null float64
High          9400 non-null float64
Low           9400 non-null float64
Close         9400 non-null float64
Volume        9400 non-null float64
Ex-Dividend   9400 non-null float64
Split Ratio   9400 non-null float64
Adj. Open     9400 non-null float64
Adj. High     9400 non-null float64
Adj. Low      9400 non-null float64
Adj. Close    9400 non-null float64
Adj. Volume   9400 non-null float64
dtypes: float64(12)
```

From this data, we can develop a linear regression model to predict the future values for APPLE's stocks.

Starting from the data frame, you need to perform the linear regression analysis and predict the stock values of a company. You can select any company for this analysis, and complete and answer the following actions:

Actions	Details
Action I. (3 marks)	Give an initial analysis of the data frame by explaining different columns. During this analysis, you need to identify the FEATURES (X) that we can use to predict the response variable/target (y) that is “ CLOSE ” (column) value for the stock on a specific date.
Action II. (2 marks)	Separate the data into Features (X) and response variable (y) .
Action III. (2 marks)	Use a scatter-plot to plot the OPEN and CLOSE values.
Action IV. (4 marks)	Find the correlations between each identified feature with CLOSE and determine that these are positive or negative.
Action V. (4 marks)	Split the data into two parts in a ratio of 0.7 for training and 0.3 for testing.
Action VI. (5 marks)	Fit a multivariate linear regression model on the training data using all selected features.
Action VII. (6 marks)	What are the intercept (θ_0) and coefficients ($\theta_1, \theta_2, \dots, \theta_n$) of the model?
Action VIII. (4 marks)	What is the R^2 score for the training data and for the test data?
Action IX. (4 marks)	Create sets of imaginary values for the FEATURES, at least 3 values sets, and predict the CLOSE value for these imaginary FEATURES values. Report your predicted values.
Action X. (6 marks)	Re-instantiate your linear regression model with the parameter “fit_intercept” set to “FALSE” and rerun you analysis on the entire features data (X). What are coefficients ($\theta_1, \theta_2, \dots, \theta_n$) of the model?
Total: (40 marks)	